

Technische Hochschule Ingolstadt
Fakultät Elektro- und Informationstechnik

Sensor Network Technologies and Sensor Fusion

Particle Filters – the Details

Joint Probability Data Association Filters

Dec 11, 2025

Welcome to the lecture!

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Sensor Networks and Sensor Fusion

Organizational Matters

- **Please register for the lecture on Moodle!**
  - Link: https://moodle.thi.de/course/view.php?id=10580
  - Key: **SensNT+DF#WS25**
- **News on Moodle:**
  - Slides for the lecture (upcoming lectures marked as WIP and are subject to change!)
  - Formulary, Exercises (Solutions provided after discussion in the lecture)
  - Exercises are optional, one task / sheet per week, but are highly recommended.
    They can be handed in for correction.
  - I will *try* to publish exercises before Wednesday‘s lecture so that initial questions can be discussed on Thursday.
- **Exam: Written exam, 90 minutes**
- **Consultation hours: Please arrange via e-mail to bjoern.giesler@thi.de.**
- **Please also don‘t hesitate to send questions, lecture feedback, change requests, etc.!**

- **In this lecture:**
  - Mathematical details of Particle Filters
  - Joint Probability Data Association Filters
  - JPDAs and Particle Filters

- **The rest of the lecture series will deal with**
  - Simultaneous Localization and Mapping
    - Map creation
    - Loop closure
    - "Hidden Robot" problem
    - Rao-Blackwellized Particle Filters
  - (DNN-based) early fusion approaches.

Visualization, and some more Examples







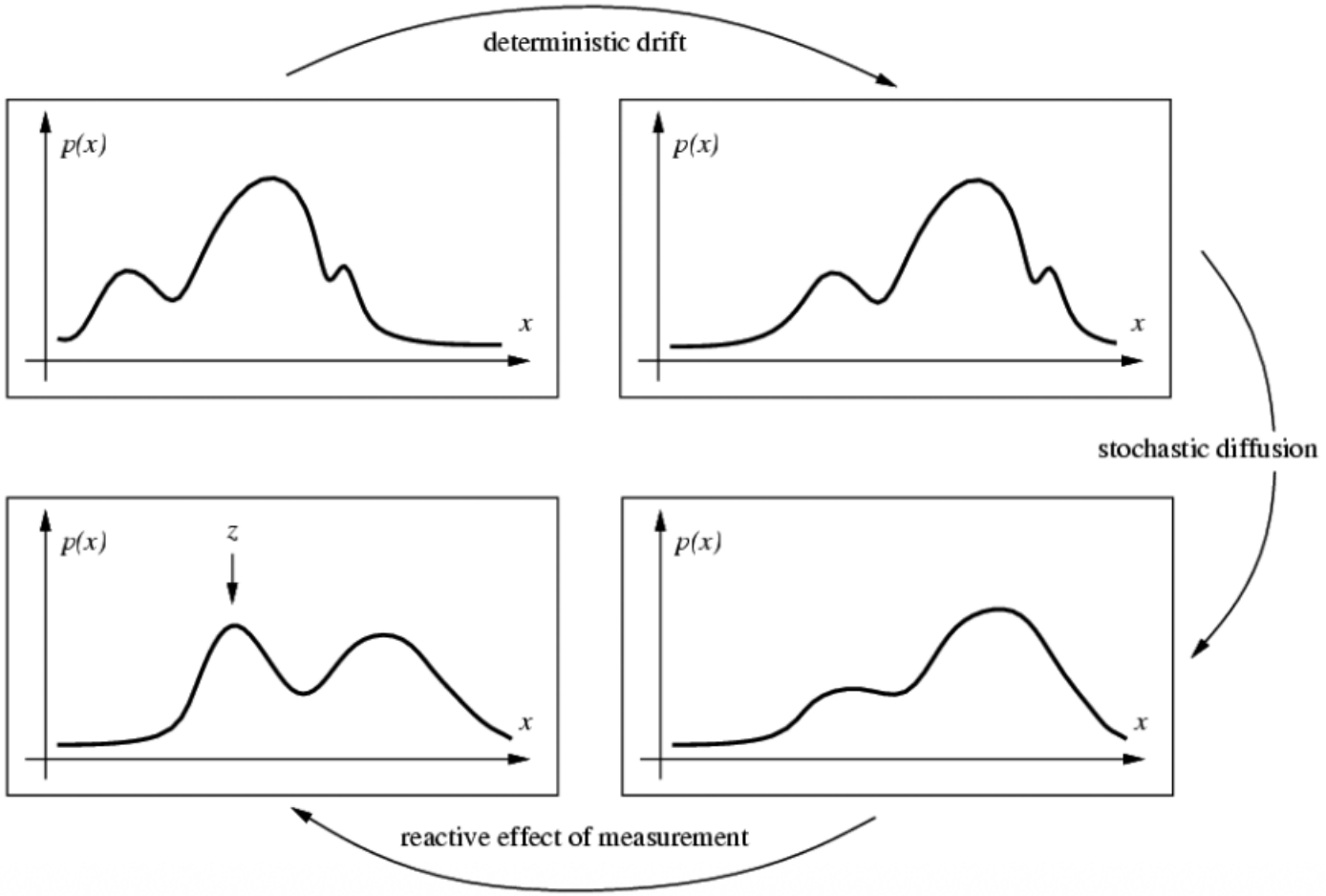


Particle Filters – the Details

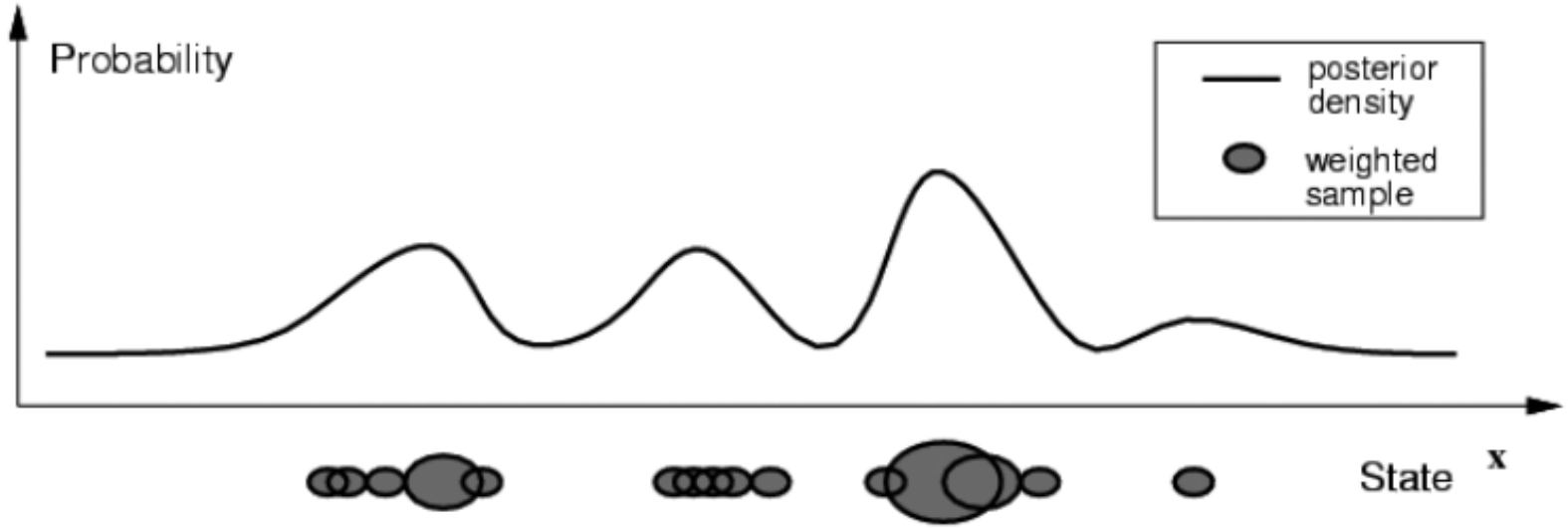
Particle Filters: Detailed Derivation

- Importance Sampling
- Sequential Importance Sampling (SIS)
- Transitional prior
- Resampling
- Generic Particle Filter
- Sampling Importance Resampling (SIR)

Propagation of General Probability Distribution Functions



Factored Sampling

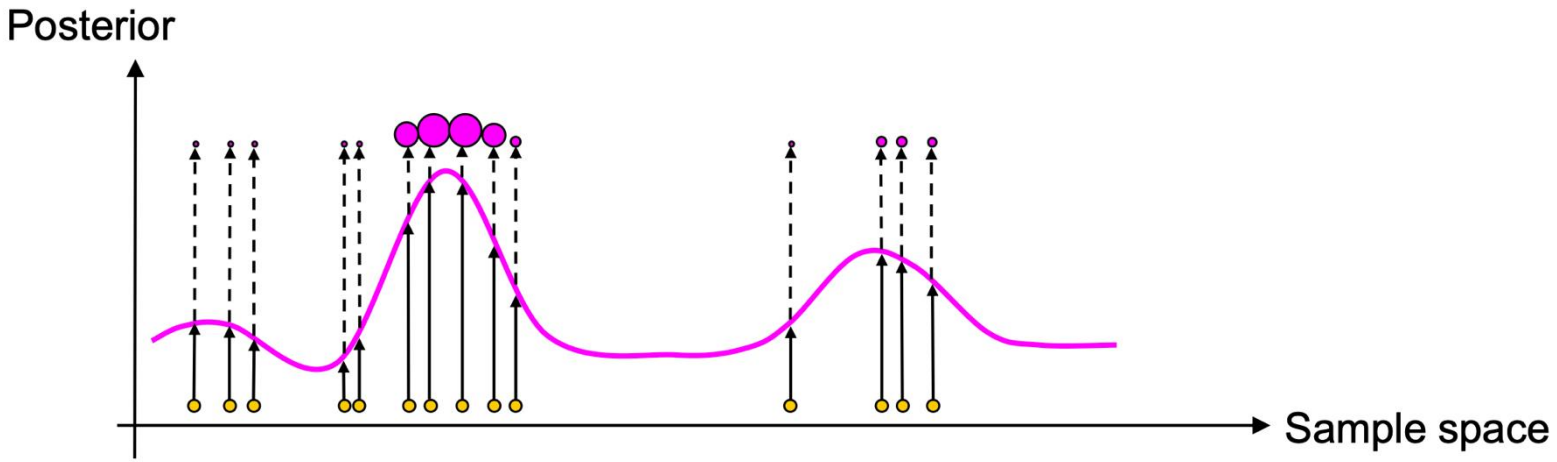


- **Idea: Represent state distribution implicitly, without explicitly modelling its parameters.**
  - Prediction: Sample points from prior density for the state, $P(X)$.
  - Correction: Weigh the samples according to $P(Y|X)$.

$$P(X_t|y_0,\ldots,y_t) = \frac{P(y_t|X_t)P(X_t|y_0,\ldots,y_{t-1})}{\int P(y_t|X_t)P(X_t|y_0,\ldots,y_{t-1})dX_t}$$

Representing PDF by Samples

- Many variations, but one general concept:
- Represent the posterior PDF by a set of randomly chosen weighted samples ("particles")



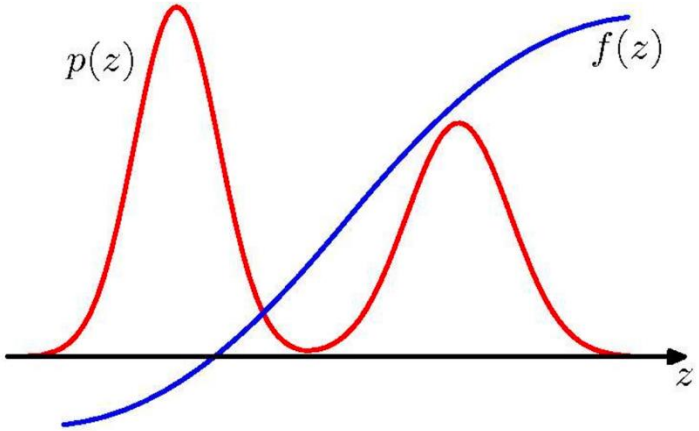
- Randomly chosen = Monte Carlo
- As the number of samples becomes very large,
  the characterization becomes an equivalent representation of the true PDF.

Monte Carlo Sampling

- **Objective:**
  - Evaluate *expectation* of a function $f(\mathbf{z})$

    regarding a probability distribution $p(\mathbf{z})$:

$$\mathbb{E}[f] = \int f(\mathbf{z})p(\mathbf{z})d\mathbf{z}.$$



  - Monte Carlo Sampling idea:
    - Draw L independent samples $\mathbf{z}^{(l)}$ with $l = 1, \ldots, L$ from $p(\mathbf{z})$.
    - Approximate expectation with the finite sum $\hat{f} = \frac{1}{L}\sum_{l=1}^{L} f(\mathbf{z}^{(l)})$.
    - As long as the samples $\mathbf{z}^{(l)}$ do not depend on each other, but are independently drawn, $\mathbb{E}[\hat{f}] \approx \mathbb{E}[f]$.

    ➔ Unbiased estimate!

Using the same idea for computing integrals

- Assume we are trying to estimate the integral of a complicated function f over some domain D:

$$F = \int_D f(\mathbf{x}) d\mathbf{x}$$

- Also assume there is an (unknown) PDF p defined over D. Then

$$F = \int_D f(\mathbf{x}) d\mathbf{x} = \int_D \frac{f(\mathbf{x})}{p(\mathbf{x})} p(\mathbf{x}) d\mathbf{x}$$

- Recalling the definition of the expectation, $\mathbb{E}[X] = \int x \cdot p(x) dx$, * the following holds:

$$\int_D \frac{f(\mathbf{x})}{p(\mathbf{x})} p(\mathbf{x}) d\mathbf{x} = \mathbb{E}\left[\frac{f(\mathbf{x})}{p(\mathbf{x})}\right], \; x \sim p.$$

* Lecture 9, slide 27

Using the same idea for computing integrals (continued)

- Now, if we have independent and identically distributed random samples $\mathbf{x}_1, \ldots, \mathbf{x}_N$ sampled from p, we can approximate the expectation

$$\mathbb{E}\left[\frac{f(\mathbf{x})}{p(\mathbf{x})}\right]$$

by

$$F_N = \frac{1}{N}\sum_{i=1}^{N}\frac{f(\mathbf{x}_i)}{p(\mathbf{x}_i)}.$$

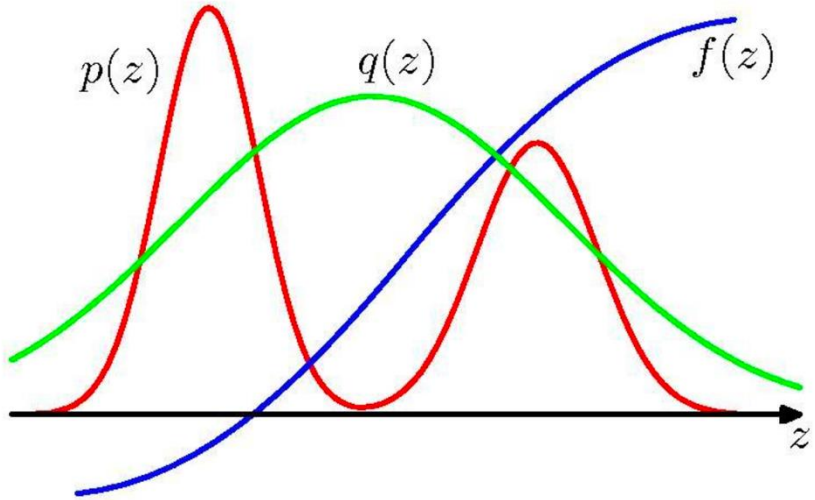
- Guaranteed by law of large numbers:

$$N \to \infty, F_N \to \mathbb{E}\left[\frac{f(\mathbf{x})}{p(\mathbf{x})}\right] = F$$

- Since it guides sampling, p is called a *proposal distribution.*

Importance Sampling

- **Let's consider an example** $F_N = \frac{1}{N} \sum_{i=1}^{N} \frac{f(\mathbf{x}_i)}{p(\mathbf{x}_i)}.$



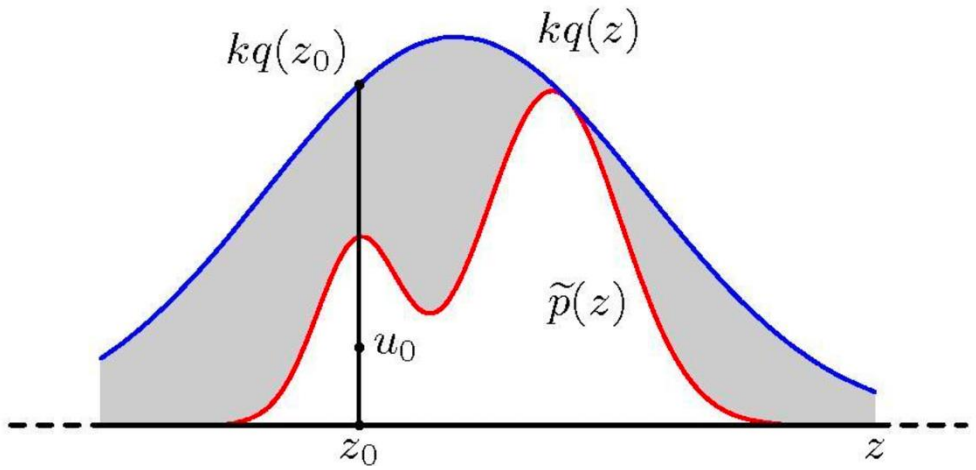
  - f/p is the *importance weight* of a sample.
  - What can go wrong here?
  - If p is very small, f/p can get arbitrarily large!
    ➔ Design p such that f/p is bounded.
  - Effect: get more samples in "important" areas of f, i.e. where f is large.

Proposal Distributions: Other Uses

- For many distributions, sampling directly from $p(\mathbf{z})$ is difficult.
- But we can often easily evaluate $p(\mathbf{z})$ (up to some normalization factor Z_p):

$$p(\mathbf{z}) = \frac{1}{Z_p}\tilde{p}(\mathbf{z})$$

- Idea:
  - Take some simpler distribution $q(\mathbf{z})$ as proposal distribution, from which we can draw samples and which is non-zero.



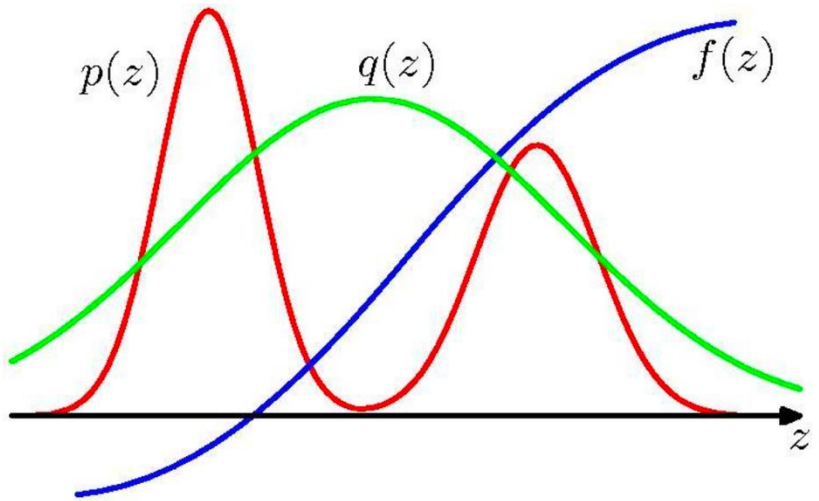
Background: Importance Sampling

- Idea: Use a proposal distribution $q(\mathbf{z})$ from which it is easy to draw samples and which is close in shape to f.
- Express expectations in the form of a finite sum over samples $\{\mathbf{z}^{(l)}\}$ drawn from $q(\mathbf{z})$.

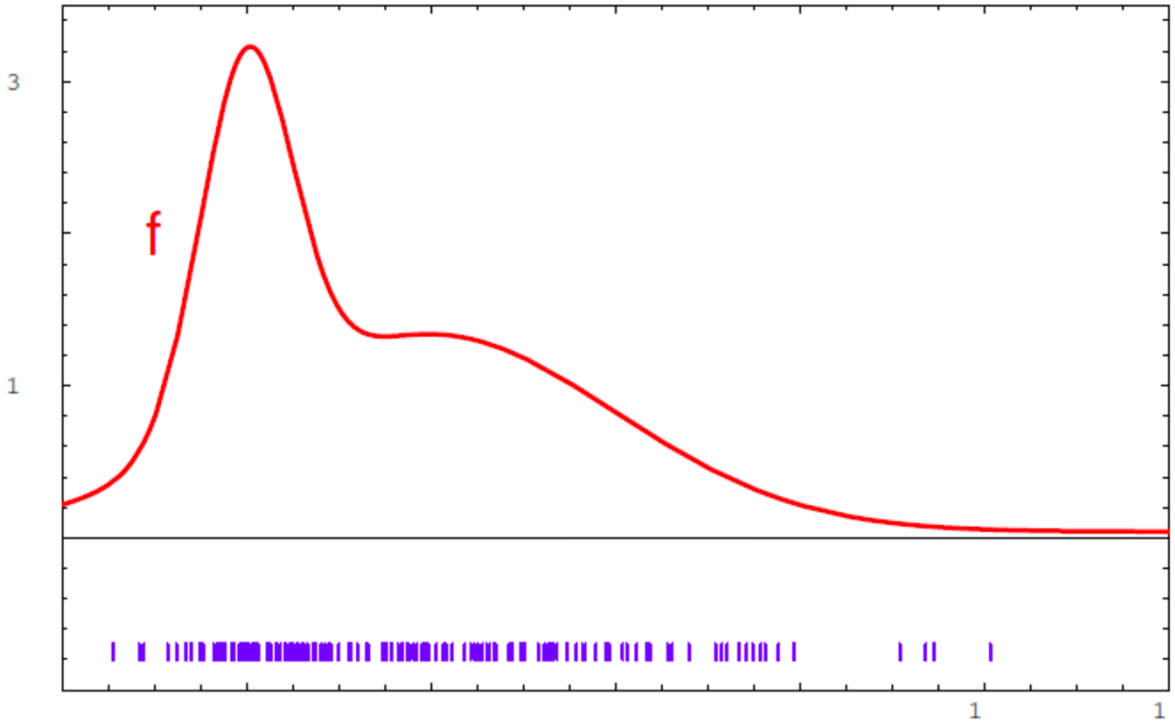
$$\begin{aligned}\mathbb{E}[f] &= \int f(\mathbf{z})p(\mathbf{z})d\mathbf{z} \\ &= \int f(\mathbf{z})\frac{p(\mathbf{z})}{q(\mathbf{z})}q(\mathbf{z})d\mathbf{z} \\ &\simeq \frac{1}{L}\sum_{l=1}^{L} f(\mathbf{z}^{(l)})\frac{p(\mathbf{z}^{(l)})}{q(\mathbf{z}^{(l)})}\end{aligned}$$

...with importance weights

$$r_l = \frac{p(\mathbf{z}^{(l)})}{q(\mathbf{z}^{(l)})}.$$

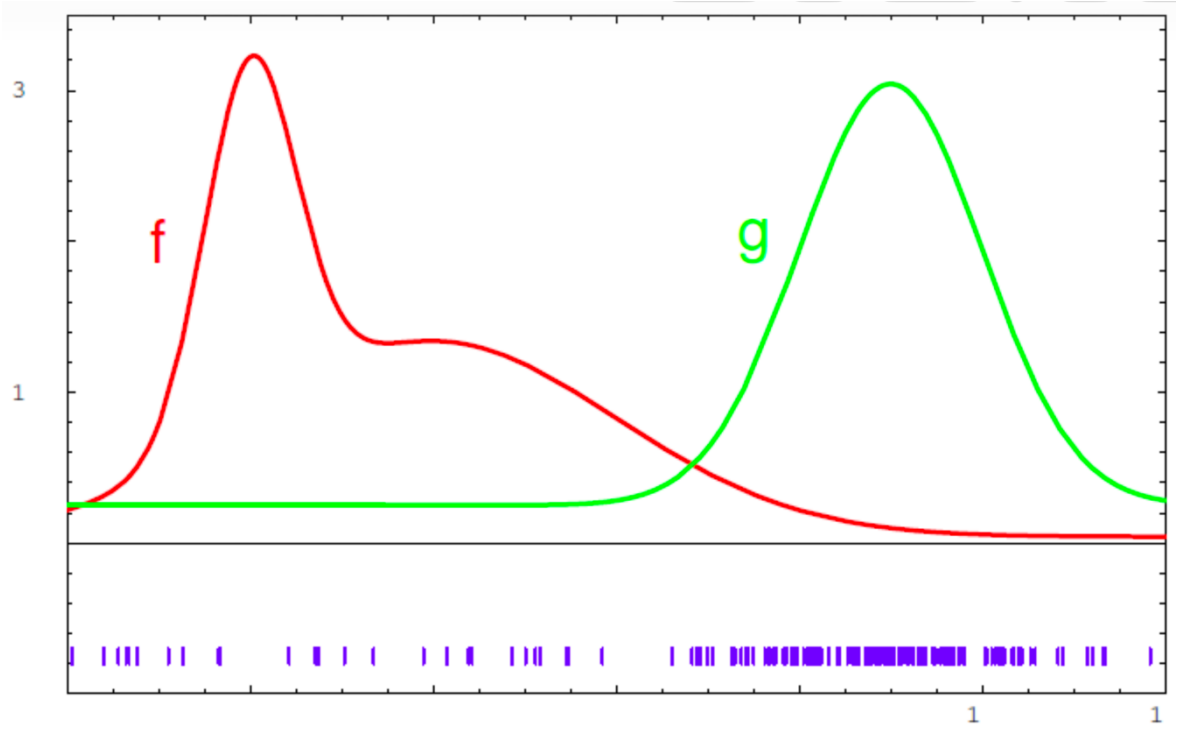


Importance Factors: Illustration



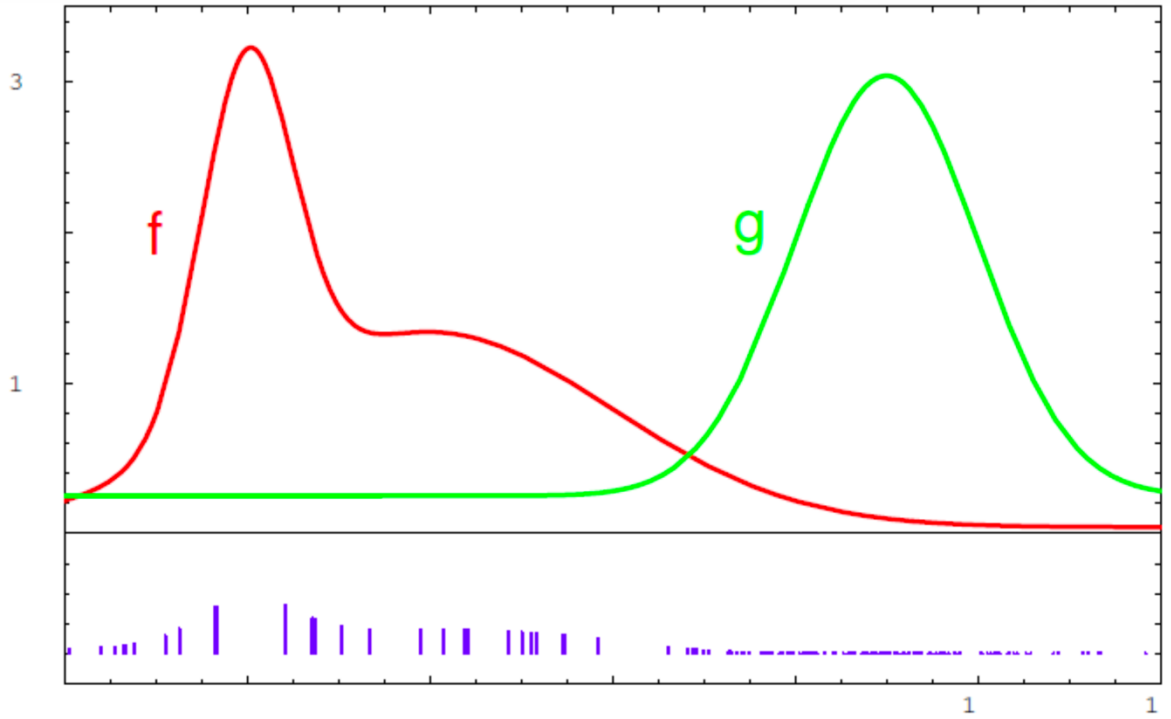
- **Goal: Approximate target density** f.

Importance Factors: Illustration



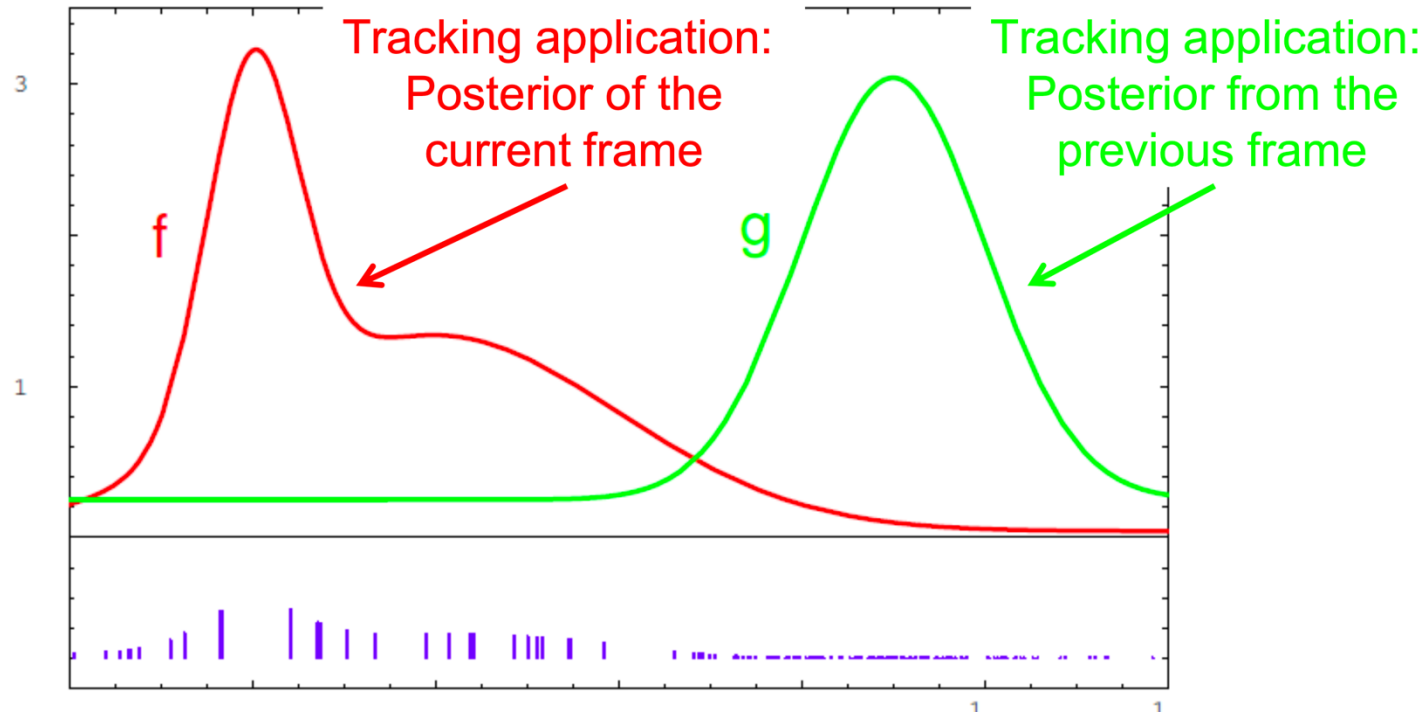
- **Goal: Approximate target density f.**
- **But instead of sampling from f directly, we can only sample from g.**

Importance Factors: Illustration



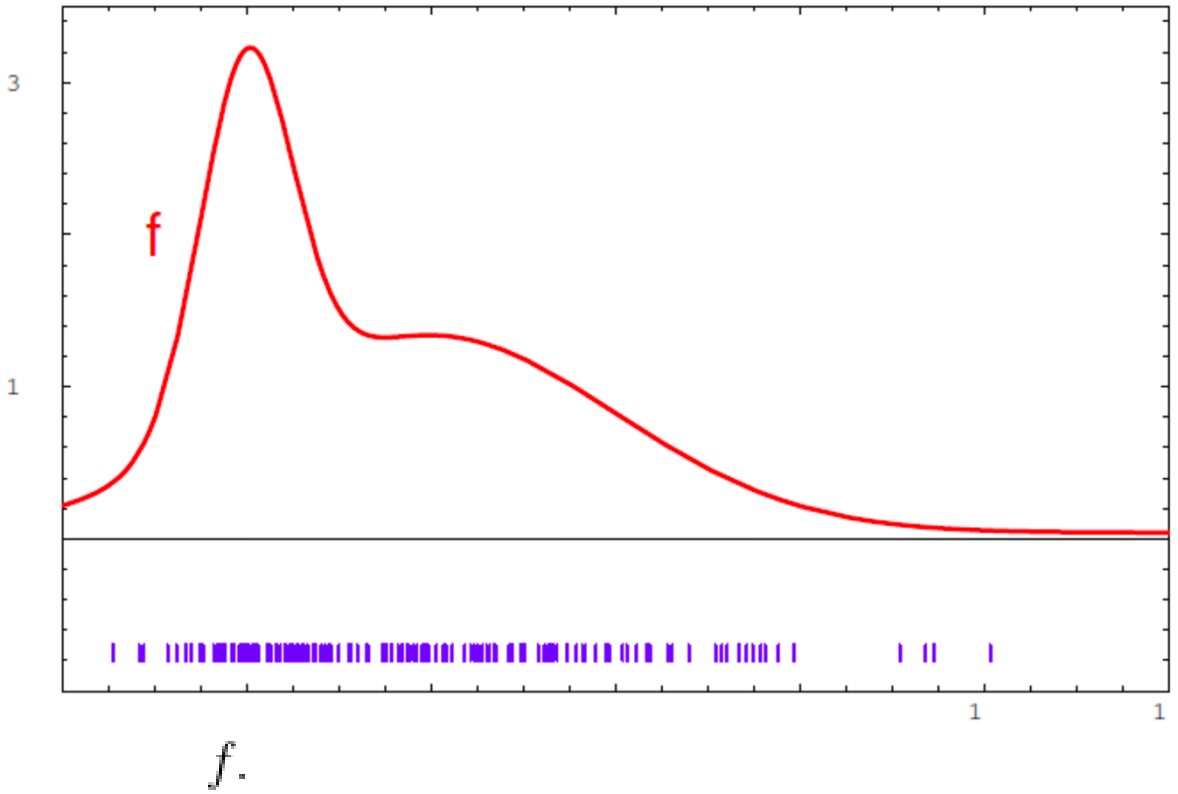
- Goal: Approximate target density f.
- But instead of sampling from f directly, we can only sample from g.
- An idea of f is obtained by attaching the weight $f(\mathbf{x})/g(\mathbf{x})$ to each sample $\mathbf{x}$.

Application



- **Goal: Approximate target density f.**
- **But instead of sampling from f directly, we can only sample from g.**
- **An idea of f is obtained by attaching the weight $f(\mathbf{x})/g(\mathbf{x})$ to each sample $\mathbf{x}$.**

Importance Factors: Illustration



Importance Sampling for Bayesian Estimation

$$\begin{aligned}\mathbb{E}[f(X)] &= \int_X f(\mathbf{x}_{0:t}) p(\mathbf{x}_{0:t}|\mathbf{y}_{1:t}) d\mathbf{x}_{0:t} \\ &= \int_X f(\mathbf{x}_{0:t}) \frac{p(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})}{q(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})} q(\mathbf{x}_{0:t}|\mathbf{y}_{1:t}) d\mathbf{x}_{0:t}\end{aligned}$$

- **Applying Importance Sampling**
  - Characterize the posterior pdf using a set of samples ("particles") and their weights

$$\{\mathbf{x}_{0:t}^i, w_t^i\}_{i=1}^N$$

  - Then we can approximate the joint posterior by

$$p(\mathbf{x}_{0:t}|\mathbf{y}_{1:t}) \approx \sum_{i=1}^{N} w_t^i \delta(\mathbf{x}_{0:t} - \mathbf{x}_{0:t}^i).$$

Importance Sampling for Bayesian Estimation

$$\mathbb{E}[f(X)] = \int_X f(\mathbf{x}_{0:t}) p(\mathbf{x}_{0:t}|\mathbf{y}_{1:t}) d\mathbf{x}_{0:t}$$
$$= \int_X f(\mathbf{x}_{0:t}) \frac{p(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})}{q(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})} q(\mathbf{x}_{0:t}|\mathbf{y}_{1:t}) d\mathbf{x}_{0:t}$$

- **Applying Importance Sampling**
  - Draw the samples from the importance density $q(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})$ with importance weights

$$w_t^i \propto \frac{p(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})}{q(\mathbf{x}_{0:t}|\mathbf{y}_{1:t})}$$

  - Particle update:

$$\mathbf{x}_t \sim q(\mathbf{x}_t|\mathbf{x}_{t-1}^i, \mathbf{y}_t)$$

  - Weight update:

$$w_t^i = w_{t-1}^i \frac{p(\mathbf{y}_t|\mathbf{x}_t^i) p(\mathbf{x}_t^i|\mathbf{x}_{t-1}^i)}{q(\mathbf{x}|x_{t-1}^i, \mathbf{y}_t)}$$

Sequential Importance Sampling Algorithm (SIS)

$$\textbf{function}\ \left[\{\mathbf{x}_t^i, w_t^i\}_{i=1}^N\right] = SIS\left[\{\mathbf{x}_{t-1}^i, w_{t-1}^i\}_{i=1}^N, \mathbf{y}_t\right]$$

Initialize

$$\eta = 0$$

$$\textbf{for}\ \ i = 1{:}N$$

Sample from proposal PDF

$$\mathbf{x}_t^i \sim q(\mathbf{x}_t|\mathbf{x}_{t-1}^i, \mathbf{y}_t)$$

Update weights

$$w_t^i = w_{t-1}^i \frac{p(\mathbf{y}_t|\mathbf{x}_t^i)p(\mathbf{x}_t^i|\mathbf{x}_{t-1}^i)}{q(\mathbf{x}_t|\mathbf{x}_{t-1}^i, \mathbf{y}_t)}$$

Update normalization factor

$$\eta = \eta + w_t^i$$

$$\textbf{end}$$

$$\textbf{for}\ \ i = 1{:}N$$

Normalize weightws

$$w_t^i = w_t^i/\eta$$

$$\textbf{end}$$

Sequential Importance Sampling Algorithm (SIS)

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Sample from proposal PDF

$$\mathbf{x}_t^i \sim q(\mathbf{x}_t|\mathbf{x}_{t-1}^i, \mathbf{y}_t)$$

Update weights

$$w_t^i = w_{t-1}^i \frac{p(\mathbf{y}_t|\mathbf{x}_t^i)p(\mathbf{x}_t^i|\mathbf{x}_{t-1}^i)}{q(\mathbf{x}_t|\mathbf{x}_{t-1}^i, \mathbf{y}_t)}$$

Need to define the importance density q!

Update normalization factor

$$\eta = \eta + w_t^i$$

$$\textbf{end}$$

$$\textbf{for}\ \ i = 1{:}N$$

Normalize weightws

$$w_t^i = w_t^i/\eta$$

$$\textbf{end}$$

Choice of Importance Density

- **Most common choice: Transitional Prior** $q(\mathbf{x}_t|\mathbf{x}_{t-1}^i, \mathbf{y}_t) = p(\mathbf{x}_t|\mathbf{x}_{t-1}^i).$

- **With this choice, the weight update reduces to**

$$\begin{aligned} w_t^i &= w_{t-1}^i \frac{p(\mathbf{y}_t|\mathbf{x}_t^i)p(\mathbf{x}_t|\mathbf{x}_{t-1}^i)}{q(\mathbf{x}_t|\mathbf{x}_{t-1}^i, \mathbf{y}_t)} \\ &= w_{t-1}^i \frac{p(\mathbf{y}_t|\mathbf{x}_t^i)\cancel{p(\mathbf{x}_t|\mathbf{x}_{t-1}^i)}}{\cancel{p(\mathbf{x}_t|\mathbf{x}_{t-1}^i)}} \\ &= w_{t-1}^i p(\mathbf{y}_t|\mathbf{x}_t^i). \end{aligned}$$

Sequential Importance Sampling with Transitional Prior

function $\left[\{\mathbf{x}_t^i, w_t^i\}_{i=1}^N\right] = SIS\left[\{\mathbf{x}_{t-1}^i, w_{t-1}^i\}_{i=1}^N, \mathbf{y}_t\right]$

$\eta = 0$

for $i = 1{:}N$

$\mathbf{x}_t^i \sim p(\mathbf{x}_t|\mathbf{x}_{t-1}^i)$

$w_t^i = w_{t-1}^i p(\mathbf{y}_t|\mathbf{x}_t^i)$

$\eta = \eta + w_t^i$

end

for $i = 1{:}N$

$w_t^i = w_t^i/\eta$

end

function $\left[\{\mathbf{x}_t^i, w_t^i\}_{i=1}^N\right] = SIS\left[\{\mathbf{x}_{t-1}^i, w_{t-1}^i\}_{i=1}^N, \mathbf{y}_t\right]$

$\eta = 0$

for $i = 1{:}N$

Draw ε_t^i *from noise distribution*

$\mathbf{x}_t^i = \mathbf{g}\left(\mathbf{x}_{t-1}^i\right) + \varepsilon_t^i$

$w_t^i = w_{t-1}^i p(\mathbf{y}_t|\mathbf{x}_t^i)$

$\eta = \eta + w_t^i$

end

for $i = 1{:}N$

$w_t^i = w_t^i/\eta$

end

The Degeneracy Phenomenon

- **Unavoidable problem with SIS.**
- **After a few iterations, most particles have negligible weights.**
- **Large computational effort for updating particles with very small contribution to** $p(\mathbf{x}_t|\mathbf{y}_{1:t})$.
- **NEFF calculation!!!**

The Degeneracy Phenomenon

- **Unavoidable problem with SIS.**
- **After a few iterations, most particles have negligible weights.**
- **Large computational effort for updating particles with very small contribution to** $p(\mathbf{x}_t|\mathbf{y}_{1:t})$.
- **Idea:**
  - Eliminate particles with low importance weights and increase the number of particles with high importance weight.
  - The *number of particles*, not their *weights*, represent the a posteriori density; so the weights are all set to uniform.

$$\{\mathbf{x}_t^i, w_t^i\}_{i=1}^N \rightarrow \left\{\mathbf{x}_t^{i*}, \frac{1}{N}\right\}_{i=1}^N.$$

  - The new set is generated by sampling with replacement from the discrete representation of $p(\mathbf{x}_t|\mathbf{y}_{1:t})$. The weights are used for multiplication of the particles:

$$P\{\mathbf{x}_t^{i*} = \mathbf{x}_t^j\} = w_t^j.$$

Resampling in Practice

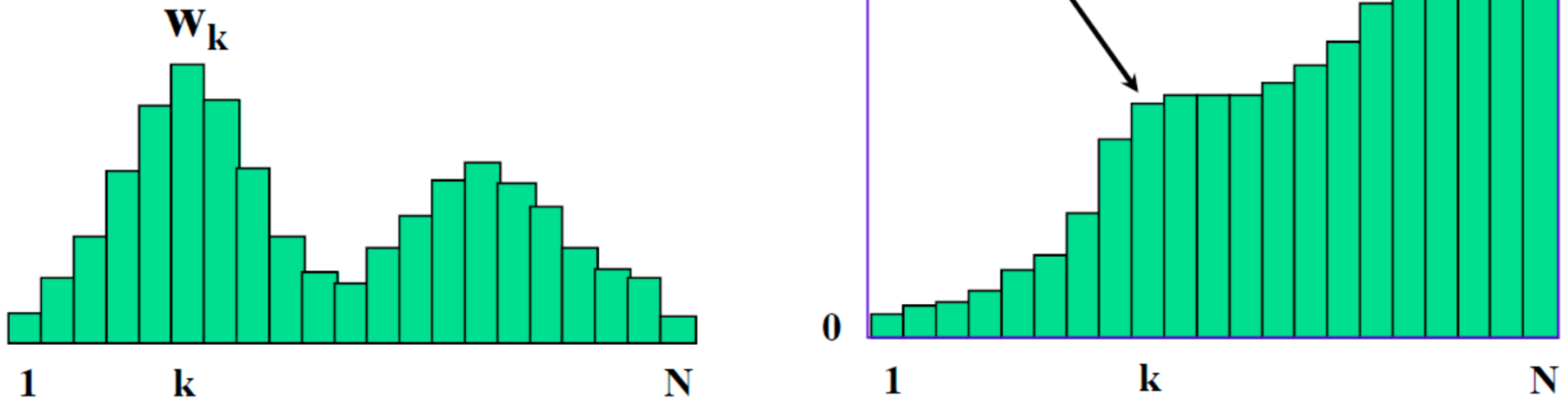
- We want to resample $\{\mathbf{x}_t^i\}_{i=1}^N$ from the discrete PDF given by the weighted samples $\{\mathbf{x}_t^i, w_t^i\}_{i=1}^N$.
- I.e. we want to draw N new samples $\{\mathbf{x}_t^i\}_{i=1}^N$ with replacement

  where the probability of drawing $\mathbf{x}_t^j$ is given by w_t^j.
- There are many algorithms for this – let's look at two simple ones.

Inverse Transform Sampling

- **Idea:**
  - It is easy to sample from a discrete distribution using the cumulative distribution function $F(x) = p(X \leq x)$.

$$c(k) = \frac{\sum_{i=1}^{k} w_i}{\sum_{i=1}^{N} w_i}$$

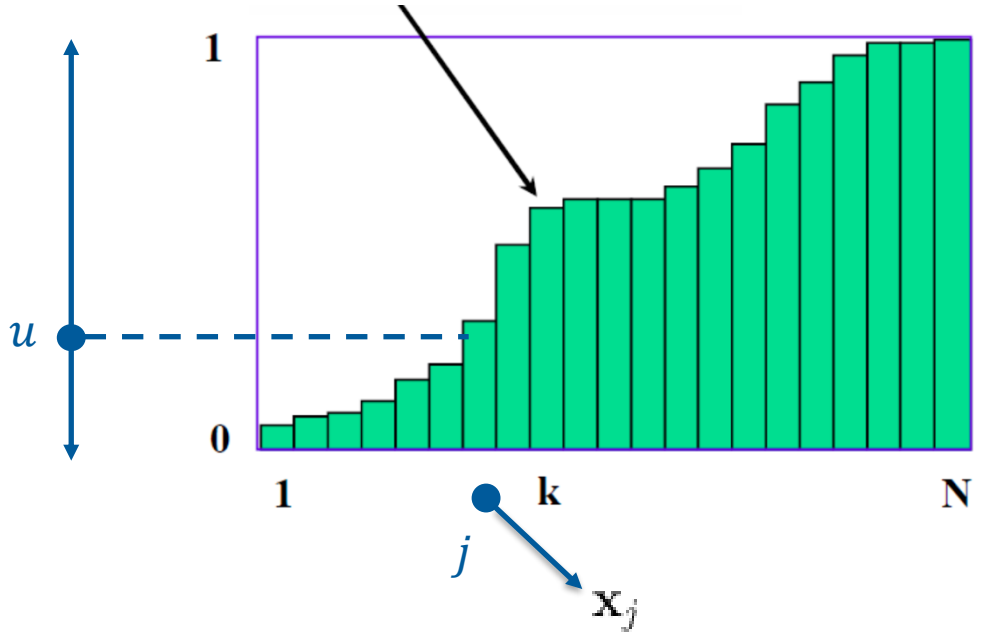


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$$c(k) = \frac{\sum_{i=1}^{k} w_i}{\sum_{i=1}^{N} w_i}$$

- **Procedure:**
  1. Generate N samples for u in the range [0,1] according to uniform distribution.
  2. Visualize a horizontal line intersecting the bars.
  3. If index of intersected bar is j, output new sample $\mathbf{x}_j$.



More Efficient Algorithm – from [1]

Basic idea:

- Choose one initial small random number
- Deterministically sample the rest by crawling up the conditional density function.
- This is O(N)!

Algorithm 2: Resampling Algorithm

$[\{\mathbf{x}_k^{j*}, w_k^j, i^j\}_{j=1}^{N_s}] = \text{RESAMPLE}\ [\{\mathbf{x}_k^i, w_k^i\}_{i=1}^{N_s}]$

- Initialize the CDF: $c_1 = 0$
- FOR $i = 2: N_s$
  - Construct CDF: $c_i = c_{i-1} + w_k^i$
- END FOR
- Start at the bottom of the CDF: $i = 1$
- Draw a starting point: $u_1 \sim \mathbb{U}[0, N_s^{-1}]$
- FOR $j = 1: N_s$
  - Move along the CDF: $u_j = u_1 + N_s^{-1}(j-1)$
  - WHILE $u_j > c_i$
    - $i = i + 1$
  - END WHILE
  - Assign sample: $\mathbf{x}_k^{j*} = \mathbf{x}_k^i$
  - Assign weight: $w_k^j = N_s^{-1}$
  - Assign parent: $i^j = i$
- END FOR

Generic Particle Filter Algorithm

$$\textbf{function} \left[\left\{\mathbf{x}_t^i, w_t^i\right\}_{i=1}^N\right] = PF\left[\left\{\mathbf{x}_{t-1}^i, w_{t-1}^i\right\}_{i=1}^N, \mathbf{y}_t\right]$$

$$Apply\ SIS\ filtering \quad \left[\left\{\mathbf{x}_t^i, w_t^i\right\}_{i=1}^N\right] = SIS\left[\left\{\mathbf{x}_{t-1}^i, w_{t-1}^i\right\}_{i=1}^N, \mathbf{y}_t\right]$$

$$Calculate\ N_{eff}$$

$$\textbf{if}\ \ N_{eff} < N_{thr}$$

$$\left[\left\{\mathbf{x}_t^i, w_t^i\right\}_{i=1}^N\right] = RESAMPLE\left[\left\{\mathbf{x}_t^i, w_t^i\right\}_{i=1}^N\right]$$

$$\textbf{end}$$

Selective resampling, only if needed (i.e. N_{eff} is too low).

This avoids drift when the tracked state is stationary.

Sampling-Importance-Resampling (SIR)

$$\textbf{function} \quad [\mathcal{X}_t] = SIR\,[\mathcal{X}_{t-1}, \mathbf{y}_t]$$

Initialize $\quad \bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$

$$\textbf{for} \quad i = 1{:}N$$

Generate new samples $\quad Sample \;\; \mathbf{x}_t^i \sim p(\mathbf{x}_t|\mathbf{x}_{t-1}^i)$

Update weights $\quad w_t^i = p(\mathbf{y}_t|\mathbf{x}_t^i)$

$$\textbf{end}$$

$$\textbf{for} \quad i = 1{:}N$$

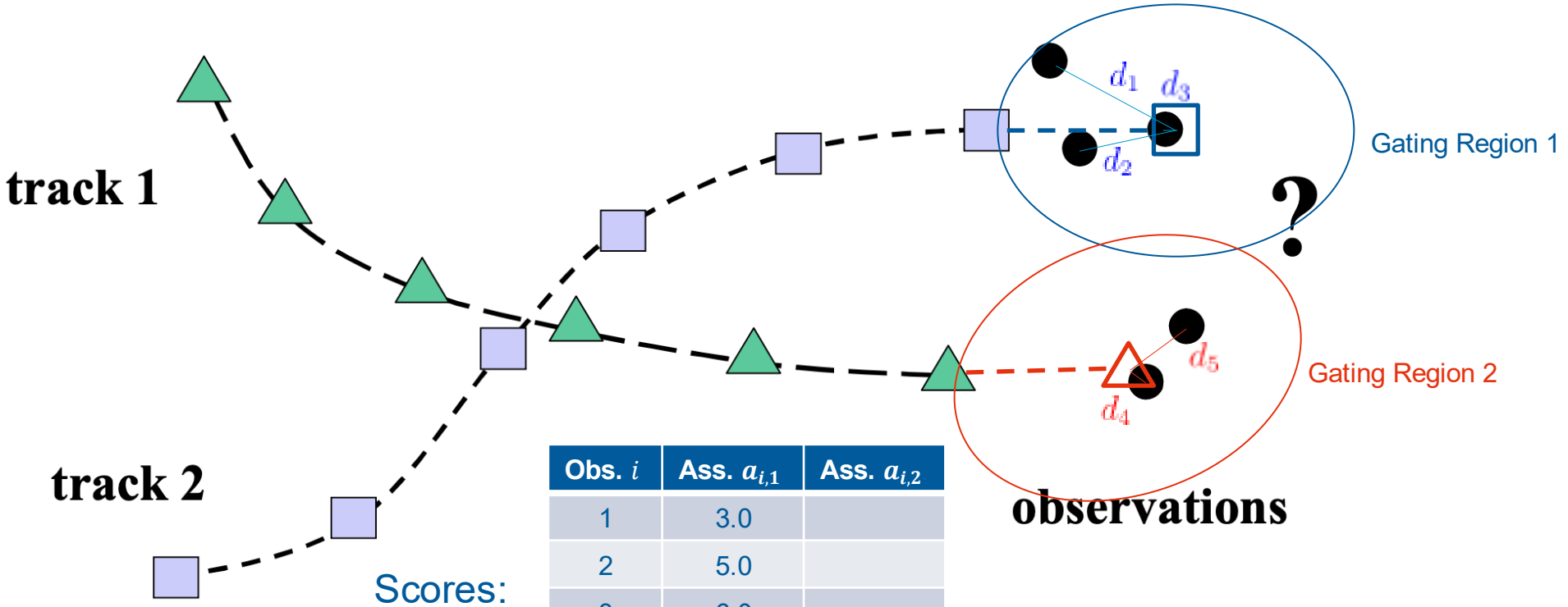
Resample $\quad Draw\ i\ with\ probability \propto w_t^i$

$$Add\ \mathbf{x}_t^i\ to\ \mathcal{X}_t$$

$$\textbf{end}$$

(Joint) Probabilistic Data Association Filter

Recall: Data Association

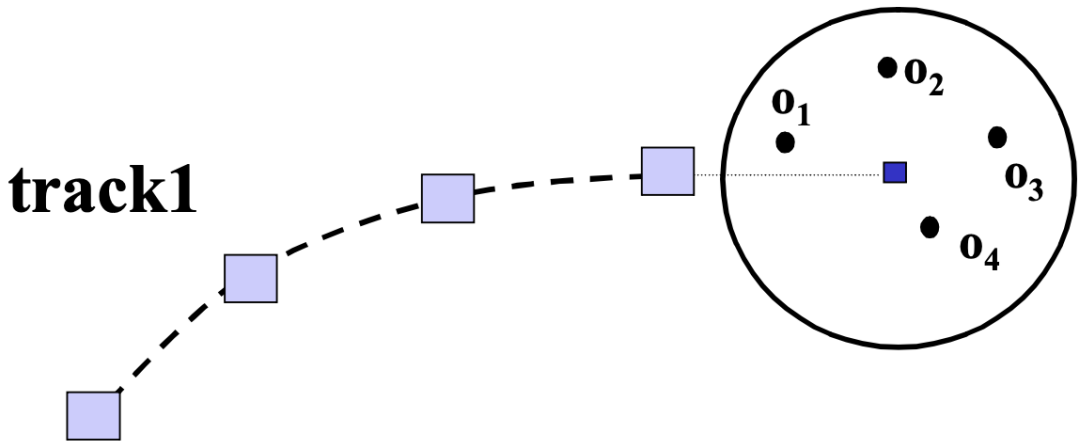


Scores:

Obs. i	Ass. $a_{i,1}$	Ass. $a_{i,2}$
1	3.0	
2	5.0	
3	6.0	
4		8.0
5		3.0

Recall: Global Nearest Neighbor

- Evaluate each observation in track gating region.
- Choose "best" one to incorporate into track.



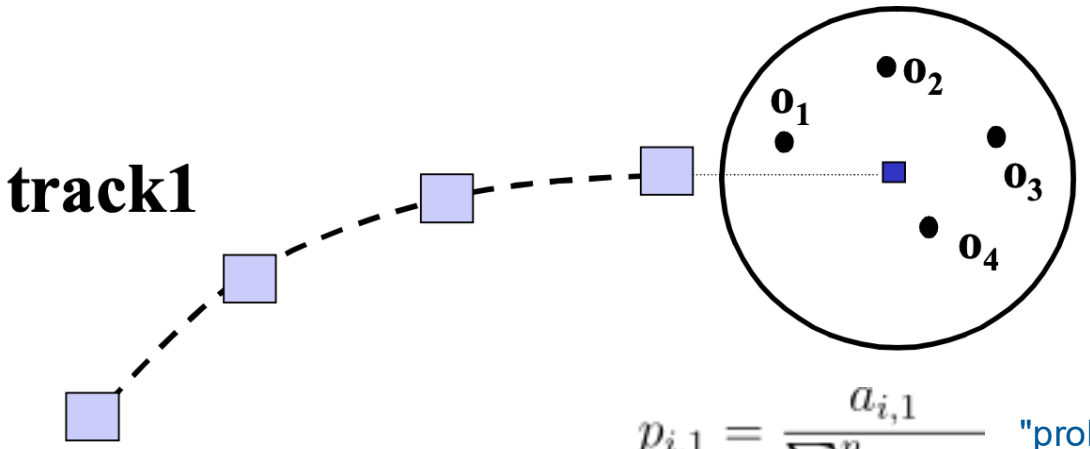
Obs. i	Ass. $a_{i,1}$	
1	3.0	
2	5.0	
3	6.0	
4	9.0	Max

$a_{i,1}$ = score for matching observation i to track 1

Choose best match $a_{m,1} = \max\{a_{1,1}, a_{2,1}, a_{3,1}, a_{4,1}\}$.

Probabilistic Data Association Filter

- Updating single track based on new observations.
- General idea: Instead of matching a single best observation to the track, we update based on all observations (in gating window), weighted by their likelihoods.
- Consider all points in gating window. Also consider the additional possibility that no observations match.



Obs. i	Ass. $a_{i,1}$
0	1.0
1	3.0
2	5.0
3	6.0
4	9.0

$$p_{i,1} = \frac{a_{i,1}}{\sum_{i=0}^{n} a_{i,1}}$$

"probability" of matching observation i to track 1

Probabilistic Data Association Filter

- $p_{i,1} = \frac{a_{i,1}}{\sum_{i=0}^{n} a_{i,1}}$ is exactly the center of mass of observations, except for the contribution of "no match".
- The Kalman filter update is based on residual vector (difference between predicted location and observed location); we called this the "innovation"
- When using the single best observation,

$$\mathbf{i}_t = \mathbf{z}_{\max} - H_t \hat{\mathbf{x}}_{t|t-1}.$$

- PDAF uses the weighted sum of the observations,

$$\mathbf{i}_t = \sum_{i=0}^{n} p_{i,1}(\mathbf{z}_i - H_t \hat{\mathbf{x}}_{t|t-1}).$$

Probabilistic Data Association Filter

- The Computation of Kalman posterior covariance must change too, to incorporate weighted matches and possibility of no match.
- Kalman filter computation when single match is used:

$$P_{t|t} = (I - K_t H_t) P_{t|t-1}.$$

- We have to change the computation when PDAF is used:

$$\begin{aligned} P_{t|t} = & \boxed{p_{0,1} P_{t|t-1}} \\ & + \boxed{(1 - p_{0,1})(I - K_t H_t) P_{t|t-1}} \\ & + \boxed{K_t \mathrm{Cov}(\mathbf{z}_i - H_t \hat{\mathbf{x}}_{t|t-1}) K_t^\top} \end{aligned}$$

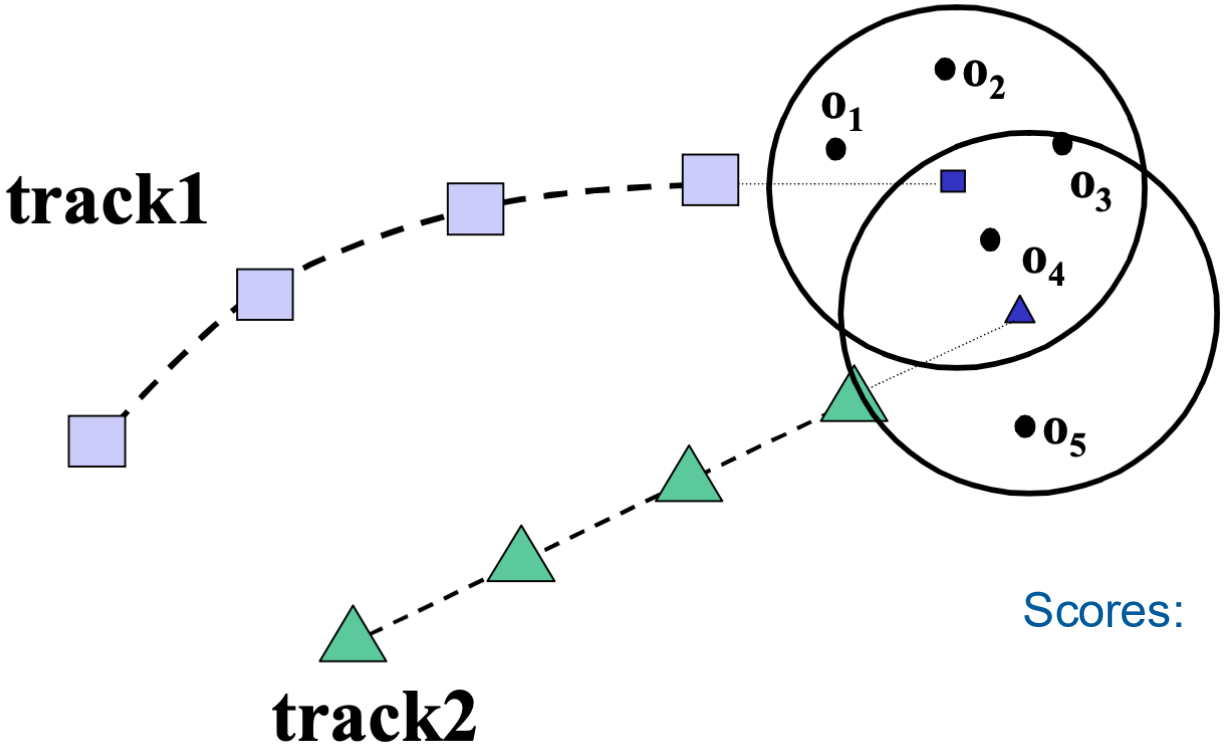
No match, no update

Update if any match

Correction term to reflect spread of possible matches

Problem

- When gating regions overlap, the same observations can contribute to updating both trajectories.
- The shared observations introduce a coupling into the decision process.

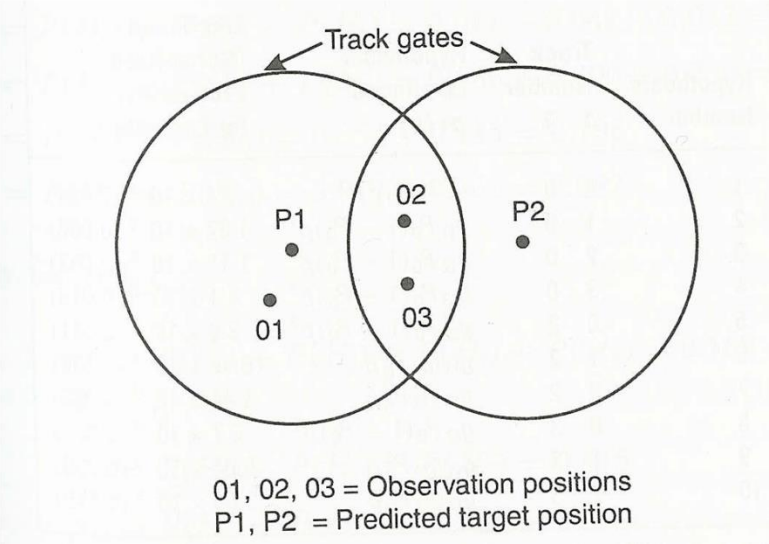


Scores:

Obs. i	Ass. $a_{i,1}$	Ass. $a_{i,2}$
1	3.0	
2	5.0	
3	6.0	1.0
4	9.0	8.0
5		3.0

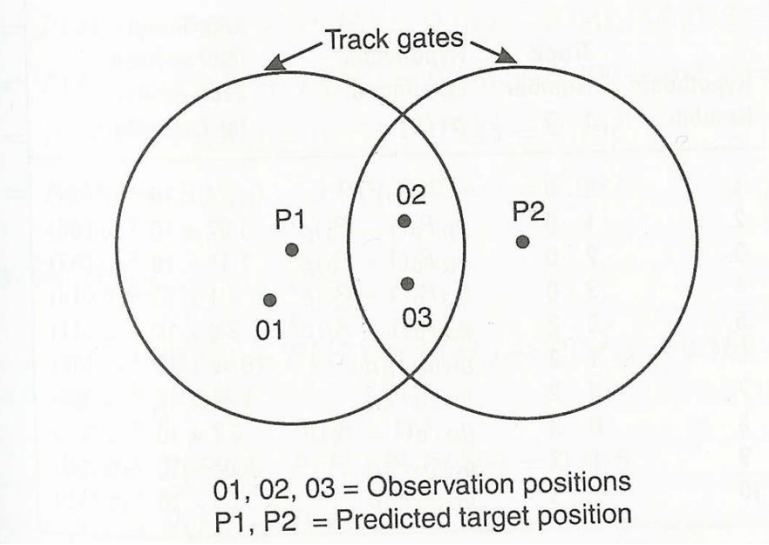
Joint Probabilistic Data Association Filter

- If maintaining multiple tracks, doing PDAF on each one independently is nonoptimal, since observations in overlapping gate regions will be counted more than once (contribute to more than one track).
- JPDAF reasons over possible combinations of matches, in a principled way.
- Following examples taken from [2].



Joint Probabilistic Data Association Filter

- **If maintaining multiple tracks, doing PDAF on each one independently is nonoptimal, since observations in overlapping gate regions will be counted more than once (contribute to more than one track).**
- **JPDAF reasons over possible combinations of matches, in a principled way.**
- **Following examples taken from [2].**



Candidates:
- Track 1: 0, 01, 02, 03
- Track 2: 0, 02, 03

Step 1: Enumerate possible assignments.
(i, j): Assign i to track 1, j to track 2

$(0,3)$ $(1,3)$ $(2,3)$ $(3,3)$ $(0,2)$ $(1,2)$ $(2,2)$ $(3,2)$ $(0,0)$ $(1,0)$ $(2,0)$ $(3,0)$

Do not assign the same observation twice

Joint Probabilistic Data Association Filter

- **Each possible (non-conflicting) assignment becomes a hypothesis with an associated probability.**

Table 6.8
Hypothesis Matrix for Example of Figure 6.3

Hypothesis Number	Track Number 1	2	Hypothesis Likelihood $p(H_i)$	Likelihood (Normalized Probability) for Example
1	0	0	$(1-P_D)^2\beta^3$	2.4×10^{-6} (0.011)
2	1	0	$g_{11}P_D(1-P_D)\beta^2$	1.82×10^{-5} (0.086)
3	2	0	$g_{12}P_D(1-P_D)\beta^2$	1.11×10^{-5} (0.053)
4	3	0	$g_{13}P_D(1-P_D)\beta^2$	4.1×10^{-6} (0.019)
5	0	2	$g_{22}P_D(1-P_D)\beta^2$	8.6×10^{-6} (0.041)
6	1	2	$g_{11}g_{22}P_D^2\beta$	6.47×10^{-5} (0.306)
7	3	2	$g_{13}g_{22}P_D^2\beta$	1.44×10^{-5} (0.068)
8	0	3	$g_{23}P_D(1-P_D)\beta^2$	6.7×10^{-6} (0.032)
9	1	3	$g_{11}g_{23}P_D^2\beta$	5.04×10^{-5} (0.239)
10	2	3	$g_{12}g_{23}P_D^2\beta$	3.06×10^{-5} (0.145)

Hypothesis likelihoods:

- P_D: Probability of a detection
- β: Probability of a false alarm
- g_{ij}: Likelihood to observe j given track i

Joint Probabilistic Data Association Filter

- Now compute probability p_{ij} that each observation i should be assigned to track j, by adding probabilities of assignments where that is so.
- Example: p_{11} = prob that observation should be assigned to track 1:

Table 6.8
Hypothesis Matrix for Example of Figure 6.3

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0.086
+ 0.306
+ 0.239
–
0.631

Joint Probabilistic Data Association Filter

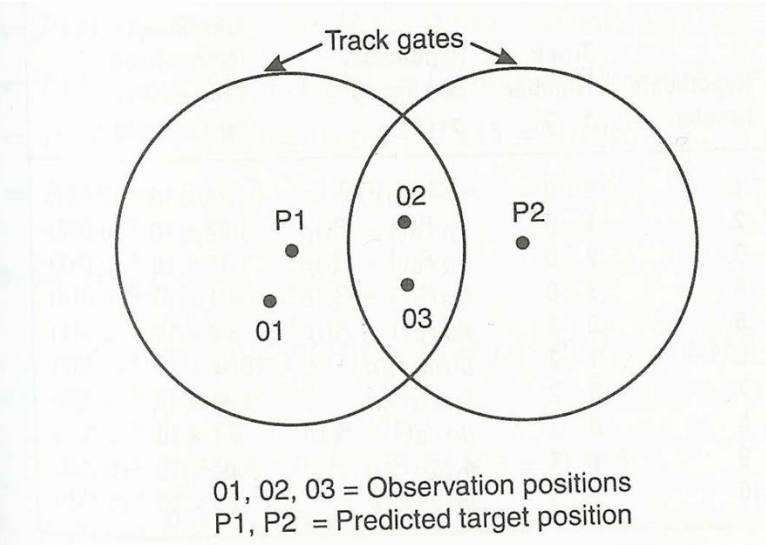


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Combi	Track 1	Track 2
p_{10}	0.084	
p_{11}	0.631	
p_{12}	0.198	
p_{13}	0.087	
p_{20}		0.169
p_{21}		0.0
p_{22}		0.415
p_{23}		0.416

PDAF filter for track 1

PDAF filter for track2

With the inconsistencies / double countings removed, we can now run the combinations into PDAF filters.

PDAF / JPDAF Summary

- **Probabilistic Data Association Filters solve the multi-observation to multi-track assignment problem by**
  - weighting gated observations by their likelihood
  - using the weighted average instead of a single measurement as the update step
  - explicitly handling the probability of "no match".
- **They need a special, but simple, adaptation for the Kalman filter to reflect the weighted average distribution.**
- **They do not handle the problem that some observations may be plausibly matched to more than one track.**
- **This can be solved by extending them with a simple elimination scheme to Joint Probabilistic Data Association Filters (JPDAFs).**

Literature / Links

1. **S.Thrun &al., *Probabilistic Robotics*, https://robots.stanford.edu/probabilistic-robotics/**
2. **S.Blackman, R.Popoli, *Design and Analysis of Modern Tracking Systems*, Artech House Radar Library 1999**



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Thank you very much!